MIL-M-81594(AS)
30 September 1968
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

MARKING FOILS; HOT STAMP PRINTING OF ELECTRICAL INSULATING MATERIALS

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes requirements for marking foils that are used in the hot stamp printing process to permanently mark the surface of electrical insulating materials (see 6.1).

1.2 Classification -

1.2.1 Types - The marking foil shall be of the following types, as specified (see 6.2).

Type I	- For neoprene
Type II	- For irradiated neoprene
Type III	- For polyethylene
Type IV	- For polyvinyl chloride
Type V	- For irradiated polyvinyl chloride
Type VI	- For nylon
Type VII	- For crosslinked polyolefin
Type VIII	- For polyethylene terephthalate
Type IX	- For polyvinylidene fluoride
Type X	- For silicone rubber
Type XI	- For fluorinated ethylene propylene (FEP)
Type XII	- For polyimide
Type XIII	- For polytetrafluoroethylene (TFE)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

SS-S-550	Sodium Chloride, Technical, for Water Softening Units
ZZ-E-661	Eraser, Rubber and Rubber Substitute
ZZ-R-765	Rubber, Silicone, Low- and High-Temperature and Tear Resistant
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall

Military

MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-W-5086	Wire, Electric, Hookup and Interconnecting, Polyvinyl Chloride - Insulated, Copper or Copper Alloy Conductor
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8243	Anti-Icing and Deicing - Defrosting Fluid
MIL-W-16878	Wire, Electrical, Insulated, High Temperature, General Specification for
MIL-I-22129	Insulation Tubing, Electrical, Polytetrafluoroethylene Resin, Nonrigid

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

Military (continued)

MIL-F-22191	Film, Transparent, Flexible, Heat Sealable, for Packaging Applications
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-W-55026	Wire, Electrical, 26 AWG, Polychloroprene Insulated
MIL-W-81044	Wire, Electric, Crosslinked Polyalkane Insulated, Copper
MIL-W-81381	Wire, Electric, Polyimide-Insulated, Copper and Copper Alloy

STANDARDS

Federal

FED-STD-601	Rubber, Sampling and Testing

Military

MIL-STD-104	Limits for Electrical Insulation Color
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-120	Gage Inspection
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MS18104	Wire, Electric, Extruded-TFE-Fluorocarbon - Insulated, Polyimide Coated, Silver-Coated Copper
MS18114	Wire, Electric, Extruded-TFE-Fluorocarbon - Insulated, Nickel-Coated Copper, 1000 Volts
MS90294	Wire, Electric, Fluorocarbon - Insulated, TFE-Glass-FEP, Medium Weight, Silver-Coated Copper, 600-Volt

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

(When requesting specifications, standards and publications refer to both the title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot (1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity, or as directed by the Contracting Officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Publications

ASTM D 1924	Practice for Determining Resistance of Plastics to Fungi

(Applications for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Data - Unless specified in the contract or order (see 6.2), no data is required by this specification or by the applicable documents referenced in Section 2.

3.2 Qualification - The marking foils furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. Each color of a specific marking foil type shall be subjected to qualification (see 4.5 and 6.3).

3.3 Properties of the marking foil -

3.3.1 Color - Unless otherwise specified by the procuring activity, a black marking foil shall be supplied for light colored insulating material and a white marking foil for dark colored insulating material. Marking colors shall conform to Class I of MIL-STD-104.

3.3.2 Dimensions and tolerances -

3.3.2.1 Foil width - Unless otherwise specified by the procuring activity (see 6.2), the marking foil shall be supplied in widths of 1/2, 3/4, 1, 1-1/4, 1-1/2, 2 and 3 inches. The width tolerance shall be +1.0, -0 percent.

3.3.2.2 Roll length - The roll length shall be as specified by the procuring activity (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-23T05:46Z
Check the source to verify that this is the current version before use.

3.3.3 Tensile strength - When tested as specified in 4.8.3, all types of marking foil shall have a minimum tensile strength of 10 pounds per square inch.

3.4 Marking application -

3.4.1 Process - The markings, to be tested as specified herein, shall be applied to the insulating material specified in 4.8.1. The hot stamping procedures employed shall be in accordance with the recommendations of the marking foil manufacturer.

3.4.1.1 Drying qualities - The marking foil shall be dry immediately after hot stamping and shall be capable of being handled at once without smudging or rubbing off. When required, Types I and X may be overcoated after hot stamping; Type XIII may be heat fused. The overcoating and heat fusion processes shall be subject to the approval of the qualifying laboratory (see 4.4).

3.4.1.2 Legibility - When examined visually at a reading distance of 14 inches; each character of the marking shall be uniform, clear, and easily identified as an accurate duplication of the printing type.

3.4.2 Printing type - The marking shall be applied with the type specified in Table I.

3.4.3 Indentation - When tested as specified in 4.8.4, the average depth of surface indentation immediately after hot stamping (except that markings on heat shrinkable sleeving shall be measured after the shrinkage and conditioning specified in 4.8.1.1) shall not exceed 0.001 inch for an insulation thickness of 0.010 inch or less, or 0.002 inch for an insulation thickness greater than 0.010 inch. But, no single measurement of indentation shall exceed 0.0015 inch for an insulation thickness of 0.010 inch or less; or 0.0025 inch for an insulation thickness greater than 0.010 inch.

3.5 Properties of the marking -

3.5.1 Electrical non-conduction - The markings, applied as specified in 3.4, shall be electrically non-conductive (see 4.4).

3.5.2 Adherence - After rubbing with an eraser as specified in 4.8.5, the marking shall conform to the legibility requirement of 3.4.1.2.

3.5.3 Fungus resistance - The markings shall not support fungus growth when tested as specified in 4.8.6.

3.5.4 Low temperature flexibility - After the low temperature flexibility test specified in 4.8.7, the markings shall conform to the legibility requirement of 3.4.1.2 and the adherence requirement of 3.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

TABLE I

PRINTING TYPE 1/

Insulation diameter 2/	Printing type: Height of type face 3/ (±.001)	Printing type: Radius of curvature of type face (±.001)	Printing type: Reading direction
For insulating materials that are not flattened during the printing operation, the following curved and flat face type shall be used:			
.035 to .039	.065	.019	Vertical
.039 to .052	.065	.024	Vertical
.050 to .060	.065	.028	Vertical
.052 to .068	.050	.028	Horizontal
.069 to .096	1/16	.038	Horizontal
.097 to .166	5/64	.055	Horizontal
.167 to .350	7/64	.125	Horizontal
.351 to .624	7/64	Flat Face	Horizontal
.625 to 1.000	1/8	Flat Face	Horizontal
over 1.000	3/16	Flat Face	Horizontal
For an insulating material that is flattened during the printing operation, the following flat face type shall be used:			
.035 to .060	.065	Flat Face	Vertical
.052 to .068	.050	Flat Face	Horizontal
.069 to .096	1/16	Flat Face	Horizontal
.097 to .166	5/64	Flat Face	Horizontal
.127 to .624	7/64	Flat Face	Horizontal
.625 to 1.000	1/8	Flat Face	Horizontal
over 1.000	3/16	Flat Face	Horizontal

1/ Dimensions in inches.
2/ For heat shrinkable sleeving, the height of the type face shall be governed by the "as supplied" insulation diameter.
3/ Other type face heights may be used when approved by the qualifying activity (see 6.3).

3.5.5 Humidity resistance - After testing as specified in 4.8.8, the markings shall conform to the legibility requirement of 3.4.1.2 and the adherence requirement of 3.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

3.5.6 Fluid resistance - After immersion in fluids as specified in 4.8.9, the markings shall conform to the legibility requirement of 3.4.1.2 and the adherence requirement of 3.5.2. After testing, the marking color shall also conform to Class I of MIL-STD-104.

3.5.7 Solvent resistance - After testing as specified in 4.8.10, the markings shall conform to the legibility requirement (3.4.1.2), adherence (3.5.2), and marking colors shall conform to Class I, MIL-STD-104.

3.5.8 Heat aging - After the heat aging and bend test specified in 4.8.11, the markings shall conform to the legibility requirement of 3.4.1.2 and the adherence requirement of 3.5.2. After testing, the marking color shall also conform to Class I of MIL-STD-104.

3.6 Shelf life - After storage as specified in 4.8.12, the marking foil shall conform to all the requirements of this specification (see 4.4 and 6.1.2).

3.7 Workmanship - When examined visually, the marking foil shall be evenly coated and free from foreign matter, holes, tears, cuts, creases, wrinkles and any other defect that would render the foil unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of the marking foil shall be classified as follows:

(a) Qualification inspection (4.5)

(b) Quality conformance inspections (4.7)

4.3 Lot formation - A lot shall consist of all the marking foil of one type, size and color that has been processed from one batch of coating compound and presented for inspection at one time. A batch shall be the quantity of coating compound that has been mixed, blended or processed in a single operation intended to make the coating compound substantially uniform. If the marking foil cannot be identified by batch, a lot shall consist of marking foil of one type, size, and color that has been manufactured by the same process, under the same conditions, at essentially the same time and presented for inspection at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

4.4 Qualification sample - The qualification sample shall consist of five rolls of the marking foil for which qualification is desired. Each roll shall be 1-1/2 inches wide and 100 feet in length. The sample shall be accompanied by a report from the manufacturer or a commercial laboratory. The report shall show the product inspection results for all the requirements of this specification and shall refer specifically to the applicable paragraphs in the specification. The report shall include an affidavit certifying that the material to be supplied conforms to the requirements for electrical non-conduction (3.5.1) and shelf life (3.6). Whenever an overcoating or heat fusion process is recommended after the hot stamping operation, a suitable sample of the overcoating material and complete processing instructions for overcoating or heat fusion shall accompany the qualification sample. The sample and reports shall be forwarded to the U.S. Naval Ammunition Depot, Crane, Indiana 47533, Attention: QECS. The sample shall be plainly identified by securely attached durable tags or labels marked with the following information:

Sample for qualification inspection

MARKING FOILS; HOT STAMP PRINTING OF ELECTRICAL INSULATING MATERIALS

Specification number

Name of manufacturer

Product code number

Date of manufacture

Submitted by (name) (date) for qualification inspection in accordance with the requirements of (give type and color) under authorization of (reference authorizing letter) (see 6.3)

4.5 Qualification inspection - The qualification inspection shall consist of an examination and testing of the qualification sample (4.4) and a review of the manufacturer's report to determine conformance to all the requirements of Section 3 of this specification. Qualification approval shall be given a marking foil that conforms to all the requirements of Section 3 of this specification.

4.6 Quality conformance samples -

4.6.1 Samples for tests - Two rolls of marking foil shall be selected at random from each lot (see 4.3).

4.6.2 Samples of the end item - A quantity of the end item shall be selected at random from each lot (see 4.3) in accordance with MIL-STD-105 and Table II. The lot size shall be the number of rolls of marking foil in the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

TABLE II

INSPECTION LEVELS - END ITEM

Inspection	Requirement	Inspection Level	A.Q.L.	Sampling Unit
Color	3.3.1	S-2	Zero defects	A roll of foil
Foil width	3.3.2.1	S-2	Zero defects	A roll of foil
Roll length	3.3.2.2	S-2	1/	A roll of foil
Workmanship	3.7	S-2	4.0	A roll of foil

1/ If the average length per roll of the selected samples is less than the length per roll specified in the contract or order, the lot represented by the sample shall be rejected.

4.6.3 Samples of preparation for delivery - Samples for the inspections specified in 4.7.3.1 and 4.7.3.2 shall be selected at random from each lot in accordance with MIL-STD-105 and Table III. The lot size shall be the number of rolls of foil for inspection 4.7.3.1 and the number of shipping containers for inspection 4.7.3.2.

TABLE III

INSPECTION LEVELS - PREPARATION FOR DELIVERY

Inspection	Inspection Level	A.Q.L.	Sampling Unit
Unit packaging (4.7.3.1)	S-3	2.5	A filled unit package
Shipping containers (4.7.3.2)	II	2.5	A shipping container fully prepared for delivery (just prior to closure)

4.7 Quality conformance inspections -

4.7.1 Inspection of samples for tests - Each roll of marking foil selected in accordance with 4.6.1 shall be used to prepare a marked specimen in accordance with 4.8.1. Each roll of marking foil shall be tested for conformance to Tensile strength (3.3.3). Each marked specimen shall be tested for conformance to Legibility (3.4.1.2), Indentation (3.4.3), Adherence (3.5.2) and Solvent resistance

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

(3.5.7). The number of determinations per sample and the reporting of results shall be in accordance with Table IV. Non-conformance of a sample of marking foil, (or of a marking obtained from the marking foil) to a single requirement shall be cause for rejection of the lot represented by the sample.

TABLE IV

REPORTING OF TEST RESULTS

Property	Requirement	Number of Determinations per Sample 1/	Results Reported As	
			Pass or Fail	Average of Multiple Determinations
Color	3.3.1	One	X	—
Foil width	3.3.2.1	Two	X	—
Roll length	3.3.2.2	One	See Note 1 of Table II	
Tensile strength	3.3.3	Three	—	X
Indentation	3.4.3	Two	(see 4.8.4)	—
Legibility	3.4.1.2	2/	X	—
Adherence	3.5.2	Two	X	—
Fungus resistance	3.5.3	Three	X	—
Low temperature flexibility	3.5.4	One	X	—
Humidity resistance	3.5.5	One	X	—
Fluid resistance	3.5.6	One	X	—
Solvent resistance	3.5.7	One	X	—
Heat aging	3.5.8	One	X	—
Shelf life	3.6	See 4.8.12	—	—
Workmanship	3.7	Two	X	—

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

TABLE IV (Continued)

1/ "Sample" refers to either the selected roll of marking foil or to the marked specimen prepared as specified in 4.8.1, as applicable. For qualification inspection only, "sample" refers to the five rolls of marking foil submitted for inspection (see 4.4).

2/ Each marking on the specimen shall be examined for conformance to 3.4.1.2.

4.7.2 Inspection of the end item - The applicable samples selected in accordance with 4.6.2 and Table II shall be examined and tested for conformance to the requirements of Table II. The number of determinations per sample and the reporting of results shall be in accordance with Table IV. If the number of defective samples exceeds the acceptable quality level (AQL) specified in Table II, the lot represented by the sample shall be rejected.

4.7.3 Inspection of preparation for delivery - The applicable samples selected in accordance with 4.6.3 and Table III shall be examined for the defects classified in 4.7.3.1 and 4.7.3.2 and to determine compliance with all the other applicable requirements of Section 5 of this specification. If the number of defective samples exceeds the acceptable quality level specified in Table III, the lot represented by the sample shall be rejected.

4.7.3.1 Unit packaging -

EXAMINE	DEFECT
Material	Not material required.
Construction	Not size, type or class specified. Closure not as specified.
Marking	Warning labels or marking missing, omitted, illegible, incorrect, incomplete. Not in accordance with contract requirements.

4.7.3.2 Shipping containers fully prepared for delivery -

EXAMINE	DEFECT
Intermediate Packaging	Not level required by contract or purchase order. Material or construction not as specified.
Packing	Not level required by contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT
Packing (Cont'd)	Materials or construction not as specified. Any nonconforming component, incomplete closures, inadequate strapping, bulged or damaged containers.
Count	Less than specified or indicated quantity of primary containers per shipping container.
Markings	Warning labels or marking missing, omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

Shipping containers fully prepared for delivery shall also be examined after closure for closure defects.

4.8 Test methods -

4.8.1 Sample preparation - Each roll of marking foil shall be conditioned for 4 hours at 25 ±2° C (75 ±4° F) and a relative humidity of 50 ±5 percent prior to testing. Unless otherwise specified, all tests shall be performed at a room temperature of 20 to 25° C (68 to 75° F). For each roll of marking foil to be tested for quality of marking a suitable length of insulating material specified in Table V shall be imprinted every two inches with a twelve character marking in accordance with the marking application requirements 3.4.1 to 3.4.3. A colored insulating material shall be used in preference to a clear or natural color whenever colors are available. The insulation color shall provide a good visual contrast to the marking foil. Heat shrinkable sleeving (MIL-I-23053) shall be marked before, and tested after the unrestricted shrinkage specified in 4.8.1.1.

4.8.1.1 Unrestricted shrinkage - Unrestricted shrinkage of marked heat shrinkable sleeving shall be in accordance with the unsupported method (4.6.5.1) of MIL-I-23053.

4.8.2 Dimensions of foil -

4.8.2.1 Length and width - Measure the foil using a ruler or standard of sufficient accuracy to ensure conformance to 3.3.2.1 and 3.3.2.2, as applicable.

4.8.3 Tensile strength - The tensile strength of marking foils shall be determined in accordance with Test Method 4111 of FED-STD-601 (see 3.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

TABLE V

MATERIALS				LOW TEMPERATURE TEST (4.8.7)					HEAT AGING TEST (4.8.11)			
FOIL TYPE	FORM 1/	REFERENCE SPECIFICATION	SIZE 2/	MANDREL OD (INCH)	WEIGHT (POUNDS)	TEMPERATURE °C (°F)	BENDING		MANDREL OD (INCH)	WEIGHT (POUNDS)	TEMPERATURE °C (°F)	TIME (HOURS)
							360 DEGREE TURNS	TURNS PER MINUTE				
I	Wire	MIL-W-55026	16	1	1	-40 ±2 (-40 ±4)	3	15	1	1	100 ±2 (212 ±4)	96
II	Sleeving	MIL-I-23053/1	(.250)	3/8	—	-55 ±2 (-67 ±4)	—	—	3/8	—	100 ±2 (212 ±4)	96
III	Wire	MIL-W-16878/10	14	2	1	-55 ±2 (-67 ±4)	3	15	3 times wire OD 3/	1	95 ±2 (203 ±4)	48
	Tubing	MIL-I-631, Type A, Sub U_a	16	3/8	—	-55 ±2 (-67 ±4)	—	—	3/8	—	100 ±2 (212 ±4)	200
IV	Wire	MIL-W-16878/1	14	2	1	-55 ±2 (-67 ±4)	3	15	3 times wire OD 3/	1	150 ±2 (302 ±4)	96
	Tubing	MIL-I-7444	16	3/8	—	-40 ±2 (-40 ±4)	—	—	3/8	—	70 ±2 (158 ±4)	96
	Tubing	MIL-I-631, Type F, Sub U_b	(.250)	3/8	—	-40 ±2 (-40 ±4)	—	—	3/8	—	100 ±2 (212 ±4)	200
V	Sleeving	MIL-I-23053/2	(.250)	5/16	—	-20 ±2 (-4 ±4)	—	—	5/16	—	130 ±2 (266 ±4)	336
VI	Wire	MIL-W-5086/1	18	3	1	-55 ±2 (-67 ±4)	1/2	30 seconds or less	4-1/2	2	120 ±1 (248 ±2)	120
VII	Sleeving	MIL-I-23053/5	(.250)	5/16	—	-55 ±2 (-67 ±4)	—	—	5/16	—	160 ±2 (320 ±4)	168
VIII	Sleeving	MIL-I-23053/7	(.250)	5/16	—	-55 ±2 (-67 ±4)	—	—	5/16	—	160 ±2 (320 ±4)	336
IX	Wire	MIL-W-81044/4	14	1	1/2	-65 ±2 (-85 ±4)	2	2	1/2	1	250 ±3 (482 ±6)	120
	Sleeving	MIL-I-23053/8	(.250)	3/8	—	-50 ±2 (-58 ±4)	—	—	3/8	—	250 ±3 (482 ±6)	168
X	Tubing	ZZ-R-765, Class 1b, Grade 60	18	3/8	—	-75 ±2 (-103 ±4)	—	—	3/8	—	225 ±3 (437 ±6)	70
	Sleeving	MIL-I-23053/10	(.250)	3/8	—	-75 ±2 (-103 ±4)	—	—	3/8	—	250 ±3 (482 ±6)	96
XI	Wire	MS90294	16	3/4	3	-65 ±2 (-85 ±4)	2	2	1/2	1	250 ±3 (482 ±6)	120
	Sleeving	MIL-I-23053/11	(.248)	3/8	—	-65 ±2 (-85 ±4)	—	—	3/8	—	250 ±3 (482 ±6)	96
	Wire	MIL-W-81381/1	14	3/8	2	-65 ±2 (-85 ±4)	2	2	3/8	2	250 ±3 (482 ±6)	120
XII	Wire	MS18104	14	3/4	3	-65 ±2 (-85 ±4)	2	2	3/4	3	275 ±3 (527 ±6)	120
XIII	Wire	MS18114	14	3/4	2	-65 ±2 (-85 ±4)	2	2	1/2	2	275 ±3 (527 ±6)	120
	Sleeving	MIL-I-23053/12, Thin Wall	(.240)	3/8	—	-65 ±2 (-85 ±4)	—	—	3/8	—	300 ±4 (582 ±8)	96
	Tubing	MIL-I-22129	14	3/8	—	-65 ±2 (-85 ±4)	—	—	3/8	—	275 ±3 (527 ±6)	96

1/ All tests shall be conducted on each referenced material.

2/ Size designations are either AWG or (I.D. (inch) as supplied).

3/ The mandrel shall be rotated to form five close 360 degree turns. Each turn shall be made within 10 seconds or less.

Source: https://assist.dla.mil -- Downloaded: 20260-23T05:46Z
Check the source to verify that this is the current version before use.

4.8.4 Indentation - The indentation into the surface of the insulating material shall be measured with a contour projector as specified in paragraph 8.5.5.1 of MIL-STD-120 or with a calibrated microscope. Measurements shall be made to an accuracy of one ten-thousandth of an inch. Three characters of the marking shall be selected at random and measured for depth of indentation immediately after the indenting operation (except that the markings on heat shrinkable sleeving shall be measured after the shrinkage specified in 4.8.1.1) and prior to heat fusion, overcoating, or other secondary bonding processes. Each measurement shall be made at the deepest point of indentation in reference to a smooth surface area immediately adjacent to the indentation. Whenever a single measurement is found to have an indentation greater than the maximum specified in 3.4.3, the marking being tested shall be considered a failure. If each measurement is less than the maximum, the three measurements shall be averaged to determine conformance of the marking to the average indentation specified in 3.4.3.

4.8.5 Adherence - The marked insulating material shall be placed on a flat, solid bench and rubbed with a flat surface of a pencil eraser conforming to the regular grade of ZZ-E-661. If necessary, insulating tubing may be supported by the insertion of a close fitting mandrel. One rub shall be made by firmly pressing the eraser on the sample and rubbing at least a 3/8 inch length of eraser along the entire length of the marking. The eraser shall be wide enough to completely cover the characters being rubbed. The maximum pressure that can be applied manually shall be used when rubbing the marking. Type XII marking foils shall be subjected to 10 rubs, all other types of foil to 20 rubs. Each eraser rub shall be made at the rate of 2 to 3 seconds. The marking shall then be examined for conformance to the legibility requirement of 3.4.1.2.

4.8.6 Fungus resistance - A two inch marked specimen shall be tested in accordance with ASTM D 1924.

4.8.7 Low temperature flexibility -

4.8.7.1 Marked wire - A twelve inch length of marked wire shall be prepared as specified in 4.8.1. One end of the wire shall be secured to the rotatable mandrel and the other end to the load weight specified in Table V. Provision shall be made for rotating the mandrel in the cold chamber by means of a control located outside the chamber. The mandrel and wire shall be conditioned for 4 hours in the cold chamber at the temperature specified in Table V. At the end of this period, the wire shall be wrapped for the number of turns at the rate of turning specified in Table V. After wrapping, the marked wire shall be conditioned to room temperature. The markings shall be examined for legibility (3.4.1.2), and then tested for adherence (3.5.2).

4.8.7.2 Marked sleeving or tubing - A six-inch length of sleeving or tubing shall be marked and conditioned as specified in 4.8.1. The marked specimen shall be placed in a cold chamber for 4 hours at the temperature specified in Table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

A steel mandrel of the size specified in Table V shall be conditioned simultaneously with the specimen. After completion of the conditioning period and while still in the chamber, the marked portion of the specimen shall be bent around the mandrel through not less than 360 degrees in 10 ±2 seconds. After bending, the specimen shall be conditioned to room temperature. The markings shall be examined for legibility (3.4.1.2), and then tested for adherence (3.5.2).

4.8.8 Humidity resistance - A 12-inch length of marked insulating material (see 4.8.1), shall be conditioned in a humidity chamber. The chamber shall be capable of maintaining an internal temperature of 71 ±2° C (160 ±4° F), and an internal relative humidity (R.H.) of 95 ±5 percent. This condition shall be maintained for a period of 6 hours. At the end of the 6-hour period, the temperature shall be lowered to 38° C (100° F) or below. After 16 hours at 38° C (100° F), heat shall be again supplied for an additional 2-hour period to stabilize the chamber at 71 ±2° C and 95 ±5 percent R.H. This cycle shall be repeated a sufficient number of times to extend the total time of the test to 360 hours (15 cycles). Condition to room temperature, examine the markings for conformance to legibility (3.4.1.2) and then test for adherence (3.5.2).

4.8.9 Fluid resistance - Twelve-inch lengths of marked insulation material as applicable (see 4.8.1), shall be immersed separately to within 3 inches of their ends in each of the fluids specified in Table VI. The volume of each fluid shall be not less than 20 times that of the specimen. After 20 hours immersion at the temperature specified in Table VI, the specimens shall be removed and excess fluid removed by blotting with a clean dry cloth. Examine the markings for legibility (3.4.1.2), and then test for adherence (3.5.2).

TABLE VI

TEST FLUIDS

Fluid	Specification	Temperature
Hydraulic fluid, petroleum base	MIL-H-5606	49 ±1° C
JP-5 fuel	MIL-T-5624	49 ±1° C
Lubricating oil	MIL-L-7808	49 ±1° C
Lubricating oil	MIL-L-23699	49 ±1° C
5% sodium chloride solution	SS-S-550	21 ±1° C
Deicing fluid	MIL-A-8243	21 ±1° C

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

4.8.10 Solvent resistance - Three suitable lengths of each insulating material shall be marked and conditioned as specified in 4.8.1. Testing shall be in accordance with Test Method 215 of MIL-STD-202, using one marked specimen for each solvent solution. After testing, the markings shall be examined for conformance to 3.3.1, 3.4.1.2, and 3.5.2.

4.8.11 Heat aging - A twelve-inch length of marked wire or a six-inch length of sleeving or tubing (as applicable) shall be prepared as specified in 4.8.1. The central portion of the wire shall then be bent at least halfway around a cylindrical mandrel of the size specified in Table V. In order to maintain a tension against the mandrel, each end of the wire specimen shall be loaded with the weight specified in Table V. The wire and mandrel or the sleeving specimen, as applicable, shall then be placed in an oven at the temperature and for the time specified in Table V. After removal from the oven, the specimen shall be cooled to room temperature within one hour. When cooled, the wire specimen shall be freed from tension, removed from the mandrel and straightened. The marking color shall be examined for conformance to Class 1 of MIL-STD-104. The specimen shall then be subjected to the applicable bend test specified in 4.8.11.1 at room temperature.

4.8.11.1 Bend test -

4.8.11.1.1 Marked wire - One end of the wire shall be secured to the mandrel and the other end to the weight specified in Table V. Unless otherwise specified in Table V, the mandrel shall be rotated until the full length of the wire is wrapped around the mandrel and is under the specified tension with adjoining coils in contact. The mandrel shall then be rotated in the reverse direction until the full length of the wire which was outside during the first wrap is now next to the mandrel. Each 360 degree turn shall be made within 10 seconds or less. This procedure shall be repeated until two bends in each direction have been formed in the same section of the wire. Each 360 degree shall turn be made within 10 seconds or less. After bending, examine the markings for legibility (3.4.1.2); and then test the markings for adherence (3.5.2).

4.8.11.1.2 Marked sleeving or tubing - Bend the marked specimen through 360 degrees in 10 ±2 seconds over the mandrel specified in Table V. Examine the markings for legibility (3.4.1.2); and then test the markings for adherence (3.5.2).

4.8.12 Shelf life - The marking foil shall be packaged as specified in 5.1. Three rolls of the packaged marking foil shall then be stored at 25 ±2° C (77 ±4° F) and 50 ±5 percent relative humidity for one year. Each roll of marking foil shall then be examined and tested for conformance to all the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A only (see 6.2).

5.1.1 Level A - The marking foil, in the width and roll length specified by the procuring activity (see 6.2), shall be supplied on a core that will not distort or change shape during handling and shipment. Each roll of foil shall be packaged individually in a paperboard box conforming to PPP-B-566. Each paperboard box shall be overwrapped and sealed with transparent film conforming to MIL-F-22191 Type III.

5.2 Packing - Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A - Shipping containers shall contain marking foil of the same type, size and color and shall enclose the contents in a snug, tight-fitting manner. Shipping containers shall conform to Class weather-resistant, Compliance Symbol V3c or V3s, Style RSC of PPP-B-636 for gross weights up to 70 pounds, or to Class 2 of PPP-B-640 for gross weights up to 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification.

5.2.2 Level B - Shipping containers shall contain marking foil of the same type, size, and color, and shall enclose the contents in a snug, tight-fitting manner. Shipping containers shall conform to Class domestic, Style RSC of PPP-B-636 for gross weights up to 70 pounds or to Class 1 of PPP-B-640 for gross weights up to 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification.

5.2.3 Level C - Packages which require overpacking for acceptance by common carrier shall be packed in exterior type shipping containers in a manner which will insure safe delivery and acceptance at the point of delivery at lowest cost. Containers shall comply with the rules and regulations applicable to the selected mode of transportation.

5.3 Marking - All containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking -

5.3.1.1 All containers shall be marked with the following:

CAUTION: STORE IN A COOL DRY PLACE, AWAY FROM HEAT AND SUNLIGHT

5.3.1.2 Unit containers shall be marked with the following:

(a) Type of insulating material on which the foil shall be used and correct marking temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

(b) Use before ---------- (give month and year).

(c) Lot number.

5.3.1.3 The surface of the core which is visible after winding the marking foil shall be marked with the following:

(a) Manufacturer's designation

(b) Type of insulating material on which the foil shall be used and correct marking temperature

6. NOTES

6.1 Intended use - The marking foil is intended for use on the particular electrical insulating material specified in 1.2.1. Markings are made by heated type pressing the foil against the surface of the electrical insulating material. Markings should be applied in accordance with approved marking procedures and Military Specifications (such as, Military Specification MIL-M-81531(AS), Marking of Electrical Insulating Materials). The markings usually consist of the circuit identification codes specified in MIL-W-5088 or MIL-W-8160. Identification of each connecting wire as to its function or diagram location facilitates the manufacture of electrical equipment and subsequent maintenance.

6.1.1 Marking foil color - A black marking should be used on insulating materials that will provide a good contrast to black. A white marking foil should be used on very dark colored insulating materials that do not provide a good contrast to black. A black marking foil will usually result in a more adherent marking than white or other colors.

6.1.2 Marking foil storage - The marking foil should be stored in the shipping container with the protective overwrap intact on each package of marking foil.

The marking foil should be stored away from direct sunlight, sources of heat, chemical fumes, and excessive moisture. Prolonged storage under conditions of high relative humidity should also be avoided.

Storage and usage should be handled on a first received-first used basis. The working supply in the marking department should be kept in small closed containers. Prolonged open exposure of the foil on the working bench should be avoided.

Marking foil that has been dried out from exposure to high temperatures in storage will require a higher marking temperature than usual. Marking foil that has been exposed for long periods to excessive moisture or a high relative humidity will probably be unsuitable for use.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:46Z
Check the source to verify that this is the current version before use.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Marking foil type and color (see 1.2.1 and 3.3.1).

(c) Whether any data is required (see 3.1).

(d) Foil width and length; and number of rolls (see 3.3.2).

(e) Level A packaging only; and level of packing (see 5.1 and 5.2).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy; however, information pertaining to qualification of products may be obtained from the Commanding Officer, U.S. Naval Ammunition Depot, Crane, Indiana 47533, Attention: QECS.

6.4 Marking foils conforming to the other requirements of this specification may, at the option of the Qualifying Activity, be placed on the Qualified Products List of products qualified under MIL-M- (AS) (see 6.3) prior to completion of the shelf life test. Removal of the marking foil from the Qualified Products List will be necessary, if subsequently, the marking foil does not meet the shelf life test requirements.

6.5 Supersession data - This specification includes the requirements of QQ-T-25 and Amendment 1, dated October 20, 1959.

6.5.1 Cross reference -

Type IV marking foil is equivalent to Type I of QQ-T-25.

Type XIII marking foil is equivalent to Type III of QQ-T-25.

Type II of QQ-T-25 is not covered herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:46Z
Check the source to verify that this is the current version before use.